DO NOW — RECALL

Hexadecimal, shifts & masks

Name: _____ Class: _____ Date: _____

1. Choose a type

2. Read an unsigned byte

3. Encode a negative number

4. Add and check the range

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Boolean: True or False.

2. Model answer

64 + 16 + 8 + 2 = 90 .

3. Model answer

+18 = 00010010 . Sign magnitude: 10010010 . Two’s complement: invert → 11101101 , add 1 → 11101110 .

4. Model answer

10010000 . 112 + 32 = 144, outside −128…127. Signed overflow occurs; the stored pattern represents −112 when interpreted as an 8-bit two’s-complement integer.

By the end, you can…

• convert positive integers between hexadecimal, binary and denary;

• apply fixed-width logical shifts and explain their limits;

• use AND, OR and XOR masks to inspect, clear, set and toggle bits.

Vocabulary — know these terms

Recall the term — Hexadecimal

Recall the term — Nibble

Recall the term — Logical shift

Recall the term — Mask

Recall the term — AND

Recall the term — OR

Recall the term — XOR

Recall the term — Toggle

NEW LEARNING

Read, annotate and explain

Quick reference

Hex: A B C D E F
10 11 12 13 14 15

1 hex digit = 4 bits
bit 7 ..... bit 0
128 1

AND preserve/clear · OR set · XOR toggle

Hex is a shorter human-readable way to write the same bit pattern. It does not reduce the number of stored bits.

Hexadecimal links directly to binary

Each hexadecimal digit maps to exactly four bits because four bits have 16 possible patterns.

Logical shifts move bits

A logical shift inserts zeros into the empty positions. Bits leaving the fixed width are lost.

AND and OR masks

AND: mask 1 preserves a bit; mask 0 clears it.

OR: mask 1 sets a bit; mask 0 preserves it.

XOR toggles selected bits

XOR produces 1 when the two input bits differ. A mask bit of 1 flips the corresponding data bit; 0 leaves it unchanged.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Link denary, hexadecimal and binary

Find the hex digits

```
173 ÷ 16 = 10 remainder 13
```

The quotient gives the first hex digit; the remainder gives the second.

Convert 10 and 13 to hex

```
10 = A
13 = D

173₁₀ = AD₁₆
```

Use four bits per hex digit

```
A → 1010
D → 1101

AD₁₆ = 10101101₂
```

Return to denary

```
(10 × 16) + 13 = 173

128 + 32 + 8 + 4 + 1 = 173
```

For binary → hex, group bits from the right in fours and pad the left group with zeros if needed.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Build the answer
Reverse check: 755 ÷ 16 gives remainders 3, F, 2.

Guided 2 — CHECK

Build the answer
The third shift discards the leftmost 1, so 352 cannot fit in 8 bits.

Guided 3 — CHECK

Build the answer
The mask keeps the low nibble and clears the high nibble.

Guided 4 — CHECK

Build the answer

Guided 5 — CHECK

Build the answer
44 ÷ 4 = 11; the discarded bits are both zero.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

One hexadecimal digit represents four binary bits.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: 2. Use hexadecimal place values

Use the method from I Do to build the answer.

Method / working:

Guided 2 — TRY: 3. Trace logical shifts in a fixed byte

Use the method from I Do to build the answer.

Method / working:

Guided 3 — TRY: 4. Preserve or clear selected bits with AND

Use the method from I Do to build the answer.

Method / working:

Guided 4 — TRY: 5. Set with OR; toggle with XOR

Use the method from I Do to build the answer.

Method / working:

Guided 5 — TRY: Divide using a shift

Use the method from I Do to build the answer.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Hexadecimal → binary
Write the binary equivalent of hexadecimal B7E .
Paraphrased exam question; original value and mark retained. OCR H446/01, page 17 .

Question 2

Denary → hexadecimal
Express 188_{10} in hexadecimal.
Paraphrased exam question; original value and mark retained. OCR H446/01, page 12 .

Question 3

Show how the bases connect
Convert 11100101_{2} to hexadecimal. Show the two nibble values. [2]
Convert $3A7_{16}$ to denary. Show the place-value calculation. [2]
Convert 214_{10} to hexadecimal. Show quotient and remainder or binary grouping. [2]
Original classroom extensions of the conversion skills in Questions 1–2.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 4

Track the stored bits
Logically shift 00110100 two places right. Give the eight-bit result and its denary value. [2]
Logically shift 10100001 one place left. Give the stored byte and explain why its denary value is not double the starting value. [3]
Give the stored byte after logically shifting 00001111 one place right. [1]
Adapted from OCR June 2017 Q6(c): values changed; left-shift and precision extensions added.
Classroom marks.

Question 5

Apply each operation independently
Start with 10100101 each time. Give the result of:
AND with 00001111 . [1]
OR with 01000010 . [1]
XOR with 11110000 . [1]
Original exam-style practice aligned to PMT bitwise manipulation and masks.

Question 6

Control an eight-bit register
Bits are numbered 7…0 from left to right. Give an operation and an eight-bit mask for each independent task. Preserve all bits not mentioned.
Clear bits 7 and 6. [2]
Set bits 3 and 0. [2]
Toggle bits 5 and 2. [2]
Original exam-style application; one mark for the operation and one for the mask.